JavaScript & Interactivity

Web Design – Lecture 6

(C1WD1A/NWD011//VT2026)

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Recap

- **Last time:**
  - Flexbox and Grid for layout
  - Responsive design with media queries
  - Mobile-first approach
  - Relative units and flexible images

- **Today:** Making pages interactive

Agenda

1. JavaScript basics
2. Interactivity
3. Live Coding

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Part 1

JavaScript basics

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The three layers

- **HTML** — Structure
  - *What is this content?*
- **CSS** — Presentation
  - *How does it look?*
- **JavaScript** — Behaviour
  - *What does it do?*

What is JavaScript?

- A programming language that runs in the browser
- Makes web pages interactive and dynamic
- The only programming language browsers understand natively

- Every modern website uses JavaScript.

A brief history of JavaScript

- **1995:** Created in 10 days by Brendan Eich at Netscape
- **2000s:** AJAX enables dynamic content without page reloads
- **2009:** Node.js brings JavaScript to the server
- **Today:** JavaScript everywhere — web, mobile, desktop, servers

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Where does JavaScript go?

- External file (recommended):

```
<script src="script.js"></script>
```

- Internal script:

```
<script>
  console.log('Hello!');
</script>
```

- Place scripts at the end of body and use defer:

```
<script src="script.js" defer></script>
```

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The Console

- Your best friend for learning and debugging.
  - **Open DevTools:** F12 or right-click → Inspect
  - **Console tab:** Run JavaScript, see errors and output

```
console.log('Hello, world!');
```

Variables

- Containers for storing values:

```
let name = 'Anna';          // can be changed
const age = 25;             // cannot be changed

name = 'Erik';              // OK
age = 26;                   // Error!
```

- Use `const` by default.
- Use `let` when you need to change the values

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Data types

```
// Strings (text)
const greeting = 'Hello';
const name = "Anna";

// Numbers
const age = 25;
const price = 19.99;

// Booleans (true/false)
const isLoggedIn = true;
const hasDiscount = false;
```

Arrays

- Ordered lists of values:

```
const colours = ['red', 'green', 'blue'];

console.log(colours[0]);      // 'red' (first item)
console.log(colours.length); // 3

colours.push('yellow');       // add to end
```

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Objects

- Collections of key/value pairs:

```
const user = {
  name: 'Anna',
  age: 25,
  isAdmin: false
};

console.log(user.name);       // 'Anna'
console.log(user['age']);     // 25

user.email = 'anna@example.com';  // add property
```

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Functions

- Reusable blocks of code:

```
function greet(name) {
  return 'Hello, ' + name + '!';
}

const message = greet('Anna');
console.log(message);        // 'Hello, Anna!'
```

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Arrow functions
(Lambda-expressions)

- A shorter syntax:

```
// Traditional
function add(a, b) {
  return a + b;
}

// Arrow function
const add = (a, b) => {
  return a + b;
};

// Even shorter for simple functions
const add = (a, b) => a + b;
```

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Conditionals

- Make decisions:

```
const age = 18;

if (age >= 18) {
  console.log('Adult');
} else {
  console.log('Minor');
}
```

The DOM

Document Object Model

- The browser turns HTML into a tree of objects.
  - JavaScript can read and modify this tree.

```
document
 └──── html
       ├──── head
       └──── body
             ├──── header
             └──── main
                   └──── p
```

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Selecting elements

```
// By ID
const header = document.getElementById('main-header');

// By CSS selector (first match)
const button = document.querySelector('.submit-btn');

// By CSS selector (all matches)
const items = document.querySelectorAll('.list-item');
```

Modifying content

```
const title = document.querySelector('h1');

// Change text
title.textContent = 'New Title';

// Change HTML
title.innerHTML = 'New <em>Title</em>';
```

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Modifying styles

```
const box = document.querySelector('.box');

// Direct style changes
box.style.backgroundColor = 'blue';
box.style.padding = '20px';

// Better: toggle classes
box.classList.add('active');
box.classList.remove('hidden');
box.classList.toggle('expanded');
```

Modifying attributes

```
const link = document.querySelector('a');
const image = document.querySelector('img');

// Get attribute
console.log(link.href);

// Set attribute
link.href = 'https://example.com';
image.src = 'new-image.jpg';
image.alt = 'Description';
```

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Part 2

Interactivity

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What are events?

- Things that happen in the browser:
  - User clicks a button
  - User submits a form
  - User presses a key
  - Page finishes loading
  - Mouse moves over an element

- JavaScript can respond to these events.

Common events

Event	When it fires
`click`	Element is clicked
`submit`	Form is submitted
`keydown`	Key is pressed
`mouseover`	Mouse enters element
`mouseout`	Mouse leaves element
`focus`	Element is focused
`blur`	Element loses focus
`load`	Page or resource loads

Event listeners

- Tell JavaScript to listen for an event:

```
const button = document.querySelector('button');

button.addEventListener('click', function() {
  console.log('Button was clicked!');
});
```

- When the button is clicked, the function runs

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Event listeners

Syntax

```
// Named function
function handleClick() {
  console.log('Clicked!');
}
button.addEventListener('click', handleClick);

// Anonymous function
button.addEventListener('click', function() {
  console.log('Clicked!');
});

// Arrow function
button.addEventListener('click', () => console.log('Clicked!'));
```

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The Event object

- The handler receives information about the event:

```
button.addEventListener('click', function(event) {
  console.log(event.type);      // 'click'
  console.log(event.target);    // the button element
});
```

Preventing default behaviour

- Some elements have default actions:

```
const form = document.querySelector('form');

form.addEventListener('submit', function(event) {
  event.preventDefault(); // Stop form from submitting
  console.log('Form handled by JavaScript');
});
```

Show/hide content

JavaScript example

```
<button id="toggle-btn">Show More</button>
<div id="content" class="hidden">Hidden content here</div>

<script>
const button = document.querySelector('#toggle-btn');
const content = document.querySelector('#content');

button.addEventListener('click', () => content.classList.toggle('hidden'));
</script>

<style>
.hidden { display: none; }
</style>
```

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Simple form validation

JavaScript example

```
const form = document.querySelector('form');
const email = document.querySelector('#email');
const error = document.querySelector('.error');

form.addEventListener('submit', (event) => {
  if (!email.value.includes('@')) {
    event.preventDefault();
    error.textContent = 'Please enter a valid email';
  }
});
```

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Character counter

JavaScript example

```
const textarea = document.querySelector('textarea');
const counter = document.querySelector('.counter');
const maxLength = 200;

textarea.addEventListener('input', () => {
  const remaining = maxLength - textarea.value.length;
  counter.textContent = remaining + ' characters remaining';
});
```

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Creating elements

```
// Create a new element
const newItem = document.createElement('li');
newItem.textContent = 'New item';

// Add it to the page
const list = document.querySelector('ul');
list.appendChild(newItem);
```

Removing elements

```
const item = document.querySelector('.item');

// Remove the element
item.remove();

// Or remove from parent
item.parentElement.removeChild(item);
```

Progressive enhancement

- **Build in layers:**
  - HTML works without CSS or JavaScript
  - CSS enhances the presentation
  - JavaScript enhances the behaviour

- If JavaScript fails, the site should still work.

Unobtrusive JavaScript

- Don’t put JavaScript in HTML:

```
<!-- Bad -->
<button onclick="doSomething()">Click</button>

<!-- Good -->
<button id="btn">Click</button>

// In your .js file
document.querySelector('#btn')
  .addEventListener('click', doSomething);
```

Accessibility consideration

- **Keyboard support:**
  - Don't rely only on mouse events
- **Focus management:**
  - Keep focus logical after changes
- **Screen reader feedback:**
  - Update ARIA attributes when content changes
- **Don't break native elements:**
  - Buttons should be <button>, not

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Part 3

Live coding

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Live Coding

Let’s add JavaScript

- We'll make our course page interactive.

- **We'll build:**
  - A collapsible schedule (click week to show/hide topics)

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Part 4

Conclusion

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Key takeaways

- JavaScript adds behaviour to web pages
- The DOM is the browser's representation of HTML
- Select elements with `querySelector`
- Respond to events with `addEventListener`
- Progressive enhancement: build in layers
- Always consider accessibility

Looking ahead

- **Next:** Introduction to Back-End Development
  - How the web really works
  - Server-side code with ASP.NET Core
  - Connecting front-end to back-end
-

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Questions?

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